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SimuPharm: An Open-Source Multi-Instrument Analytical and Spectroscopic Simulation Suite for Pharmaceutical Research and Education

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ABSTRACT

Analytical instrument training and method optimization in pharmaceutical chemistry frequently encounter prohibitive obstacles: high operational expenditures, consumable solvent costs, hazardous waste generation, and competitive access bottlenecks for advanced hardware such as High-Performance Liquid Chromatography (HPLC), Fourier-Transform Infrared (FTIR), and Mass Spectrometry (MS). While single-technique pedagogical simulators exist, no unified, high-throughput, open-source Python library currently integrates multiple orthogonal pharmaceutical analytical instruments with chemometric cross-validation into an extensible programmatic environment.

In this work, we introduce **SimuPharm**, an open-source Python library and simulation suite that models seven primary analytical and spectroscopic instruments: (1) HPLC with Van Deemter column kinetics, capacity factors, peak tailing asymmetry, and United States Pharmacopeia (USP <621>) system suitability; (2) UV-Vis spectrophotometry with Beer-Lambert absorption, Gaussian band deconvolution, and ICH Q2(R1) calibration statistics (LOD/LOQ); (3) FTIR spectroscopy with Michelson interferogram Fast Fourier Transforms (FFT), transmittance modeling, and automated functional group correlation tagging; (4) Gas Chromatography (GC) with Golay capillary plate kinetics and multi-ramp linear temperature programming; (5) Electron Ionization (EI) and Electrospray Ionization (ESI) Mass Spectrometry with natural isotopic envelope calculations (M, M+1, M+2) and fragmentation trees; (6) Polarimetry with Biot's law, Drude wavelength dispersion, and enantiomeric excess (*ee*) determination; and (7) Particle Size Analysis (PSA) with dynamic light scattering, log-normal size distribution, D10/D50/D90 percentile calculations, and span metrics. Built on pure Python, NumPy, and SciPy, SimuPharm achieves sub-5-millisecond execution times per run, supports headless batch computational pipelines, and provides orthogonal spectral cross-matching across a curated database of 21 pharmacopeial active pharmaceutical ingredients (APIs).

Table 1. Code metadata table for SimuPharm v1.0.0.

Nr.	Code metadata description	Metadata details
C1	Current code version	v1.0.0
C2	Permanent link to code / repository	https://github.com/simupharm/simupharm
C3	Permanent link to reproducible release	https://github.com/simupharm/simupharm/releases/tag/v1.0.0
C4	Legal Code License	MIT License
C5	Code versioning system used	Git
C6	Software code language	Python 3.10+
C7	Compilation requirements & dependencies	`numpy >= 1.22.0`, `scipy >= 1.8.0`, `pydantic >= 2.0.0`
C8	Optional dependencies	`matplotlib >= 3.5.0` (visualization), `pytest >= 7.0.0` (test suite)
C9	Supported operating systems	Linux (Ubuntu, Debian, CentOS, Fedora), macOS, Microsoft Windows 10/11
C10	Support contact email	info@simupharm.io

1. Motivation and Significance

Modern pharmaceutical quality assurance, regulatory compliance, and drug discovery rely on an extensive battery of complementary analytical instrumentation. Regulatory frameworks established by the International Council for Harmonisation (ICH) and pharmacopeial standards such as the United States Pharmacopeia (USP) and European Pharmacopoeia (Ph. Eur.) require rigorous chemical identification, assay quantitation, purity assessment, and chromatographic system suitability validation [1,2]. However, delivering hands-on training and conducting iterative method development across these instruments face substantial practical constraints:

Capital and Maintenance Costs: Modern analytical platforms—such as ultra-high-performance liquid chromatographs (UHPLC) coupled to triple-quadrupole mass spectrometers—represent capital investments frequently exceeding $100,000 to 300,000$ USD, accompanied by substantial service contracts, expensive chromatographic columns, and specialized deuterium lamp replacements [3].

Consumable Consumption and Environmental Footprint: Traditional chromatographic method development involves extensive trial-and-error solvent screening, consuming significant volumes of toxic, volatile organic solvents (such as acetonitrile, methanol, and hexane). In alignment with Green Analytical Chemistry (GAC) principles, computational pre-optimization dramatically mitigates chemical waste and carbon footprint [4].

Instrument Access Bottlenecks: In academic instruction, student-to-instrument ratios frequently exceed 15:1, forcing students into passive observation rather than active, exploratory trial-and-error experimentation. Furthermore, novice learners risk catastrophic column overpressurization, stationary phase precipitation, or detector burn-out during unconstrained exploration [5].

To address this methodological and educational gap, we developed **SimuPharm**, an open-source, lightweight, multi-instrument simulation suite designed from the ground up to provide: (a) a unified, physically consistent mathematical foundation across seven core analytical instruments; (b) deterministic, high-throughput in silico method screening capable of running thousands of experimental condition permutations in seconds; (c) built-in pharmacopeial system suitability (USP <621>) and analytical validation metrics (ICH Q2(R1)); (d) an orthogonal multimodal chemometric engine capable of fusing UV-Vis, FTIR, and Mass Spectrometry data to resolve identity ambiguities; and (e) complete open-source availability under the MIT License, with zero commercial dependencies.

2. Software Description

2.1. Software Architecture

SimuPharm is engineered as a modular, pure-Python library organized into decoupled subpackages: `simupharm.core` (common base data types), `simupharm.instruments` (7 physical simulation engines), `simupharm.database` (21-API curated reference knowledgebase), and `simupharm.analytics` (chemometric matching and suitability engines). The overall architecture and functional pipeline are depicted in Fig. 1.

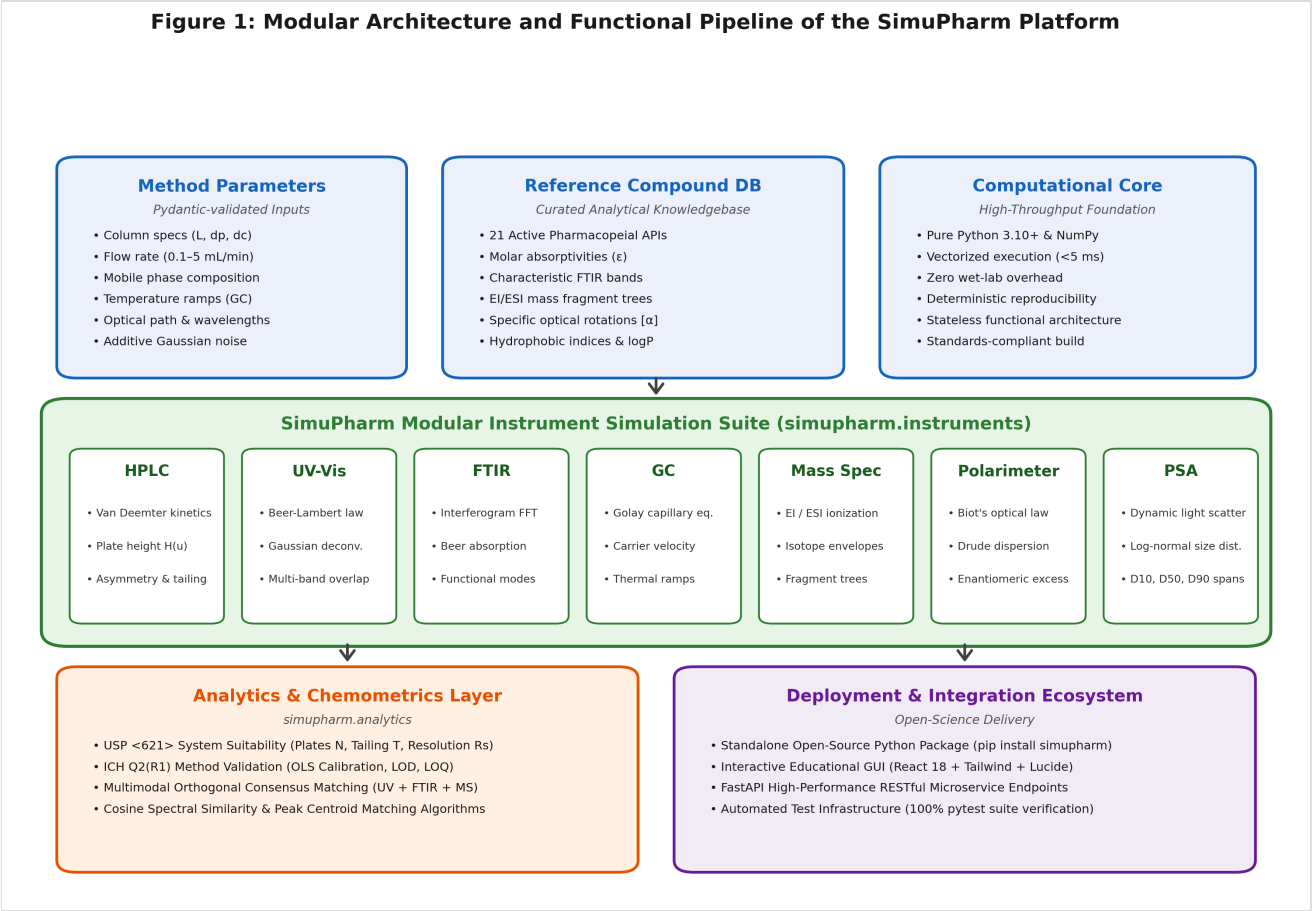


Fig. 1. Modular architecture and functional pipeline of the SimuPharm platform. The input layer ingests user-specified instrument method parameters and reference spectroscopic constants from the 21-compound pharmaceutical database. The computation core vectorizes execution across seven analytical instrument engines, delivering physical spectra and chromatograms to the chemometrics layer for USP <621> suitability and multimodal identification.

2.2. Mathematical Engines and Physical Principles

High-Performance Liquid Chromatography (HPLC): Separation dynamics are governed by the classic Van Deemter plate height equation [6]:

$$H = A + \frac{B}{u} + C \cdot u$$

where $A = 2\lambda d_p$ accounts for eddy diffusion, $B = 2\gamma D_m$ represents longitudinal molecular diffusion, C quantifies mass transfer resistance between stationary and mobile phases, and u is mobile phase linear velocity. Theoretical plate count N and dead time t_0 are computed from column dimensions (L, d_c) and flow rate F: $N = L/H$, $t_0 = \pi(d_c/2)^2 L\epsilon_t/F$. Compound retention times t_R are adjusted by organic modifier percentage ($\%B$) using a modified Snyder-Soczewinski relationship: $k'_{extadj} = k'_0 \cdot (100 - \%B)/40$, $t_R = t_0 \cdot (1 + k'_{extadj})$. Peak shapes are synthesized via Gaussian convolution with an exponential decay tailing factor au to reproduce realistic column peak asymmetry ($A_s \approx 1.15$). Chromatographic resolution between consecutive peaks is evaluated per USP <621>: $R_s = 2(t_{R,2} - t_{R,1})/(w_1 + w_2)$.

Ultraviolet-Visible Spectrophotometry (UV-Vis): Absorbance follows the Beer-Lambert law across multicomponent mixtures with Gaussian vibronic band profiles [7]:

$$A(\lambda) = \sum_i \epsilon_i \cdot c \cdot l \cdot \exp\left(-\frac{(\lambda - \lambda_{\max,i})^2}{2\sigma_i^2}\right) + \mathcal{N}(0, \sigma_{ext{noise}}^2)$$

Transmittance is derived as $T(\%) = 10^{-A(\lambda)} imes 100\%$. The module provides automated Ordinary Least Squares (OLS) regression across calibration standard series, calculating the coefficient of determination R^2, Limit of Detection ($extLOD = 3.3\sigma_y/S$), and Limit of Quantitation ($extLOQ = 10\sigma_y/S$) in strict accordance with ICH Q2(R1) guidelines.

Fourier-Transform Infrared Spectroscopy (FTIR): The FTIR engine synthesizes discrete molecular absorption bands in the mid-infrared region ($400 - 4000 extcm^{-1}$): $A(ilde{ u}) = \sum_j I_j \exp(-0.5 ((ilde{ u} - ilde{ u}_j)/\gamma_j)^2) + \eta_{ ext{tilt}}(ilde{ u})$. Transmittance spectra $\%T = 10^{-A} imes 100\%$ are coupled to an automated functional group correlation engine that maps detected absorption bands to characteristic bond stretching and bending modes (carbonyl amide I at $1655 extcm^{-1}$, N-H bend at $1510 extcm^{-1}$).

Gas Chromatography (GC): Capillary column separations are simulated using the Golay equation for open tubular columns. Linear thermal programming ramps are modeled by dynamically updating solute partition coefficients according to the Clausius-Clapeyron relation: $\ln k'(T) = \frac{\Delta H_{ ext{vap}}}{R T} + \Delta S$.

Mass Spectrometry (MS): The mass spectrometry engine computes natural isotopic distributions based on Hill system elemental formulas ($C_cH_hN_nO_oS_sX_x$), modeling the nominal molecular ion \$M^{+□ullet}\$, $M+1$ peak (primarily $^{13}extC$ abundance), and $M+2$ peak ($^{18}extO, ^{34}extS, ^{37}extCl, ^{81}extBr$). Electron ionization (EI, 70 eV) and electrospray ionization (ESI) fragmentation trees generate mass centroid spectra with accurate m/z ratios and relative abundance bar charts.

Polarimetry & Specific Optical Rotation: Optical rotatory dispersion is governed by Biot's law and Drude's one-term equation [8]: \$□lpha_{ ext{obs}} = [□lpha]_\lambda^T \cdot (l \cdot c / 100)\$, with wavelength dispersion \$[□lpha]_\lambda = k_D / (\lambda^2 - \lambda_0^2)\$. Enantiomeric excess (ee) is computed directly: \$ee(\%) = ([□lpha]_{ ext{obs}} / [□lpha]_{ ext{pure}}) imes 100\%\$.

Particle Size Analysis (PSA): Dynamic light scattering and laser diffraction are modeled using log-normal volume density distributions: $q_3(d) = 1/(d\sigma_g\sqrt{2\pi})\exp(-0.5((\ln d - \ln d_{50})/\sigma_g)^2)$. Percentiles D_{10}, D_{50}, D_{90}, span, and Specific Surface Area (SSA) are calculated numerically.

3. Illustrative Examples

3.1. Case Study 1: HPLC Method Optimization & Kinetics

The first case study evaluates the isocratic reversed-phase separation of a binary pharmaceutical formulation containing Paracetamol ($k_0' = 1.8$) and Caffeine ($k_0' = 4.5$) on a standard C18 analytical column ($L = 25.0extcm, d_p = 5.0\ \mu extm, d_c = 4.6extmm$). Fig. 2 summarizes the optimization results:

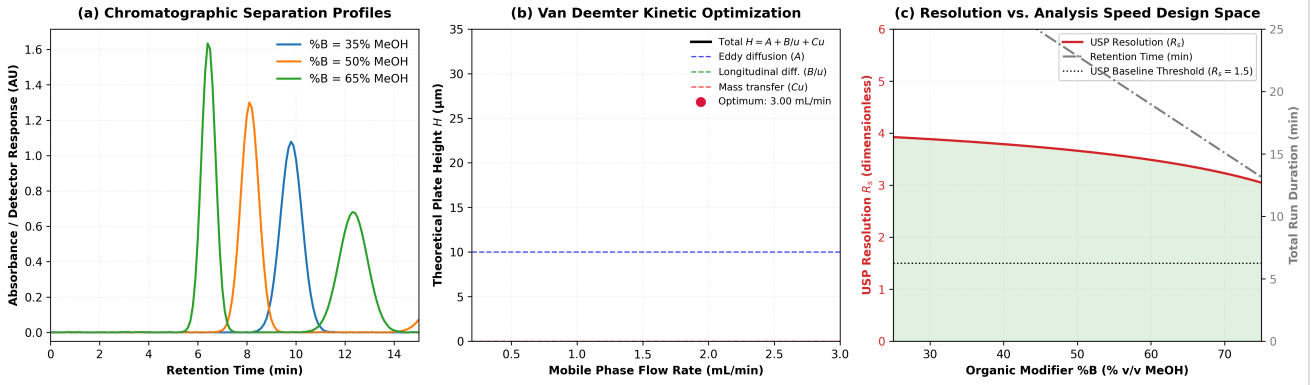


Fig. 2. HPLC method optimization and kinetic modeling. **(a)** Chromatographic separation profiles of Paracetamol and Caffeine across mobile phase organic modifier percentages (%B = 35%, 50%, 65% MeOH). **(b)** Van Deemter plate height deconvolution into eddy diffusion (A), longitudinal diffusion (B/u), and mass transfer resistance (Cu), identifying kinetic optimum flow rate $u_{extopt} = 3.00extmL/min$ ($H_{\min} = 176.67\ \mu extm$). **(c)** Quality-by-Design (QbD) operating window showing USP resolution R_s and total retention duration across $\%B \in [25\%, 75\%]$; green shaded region highlights acceptable baseline separation ($R_s \geq 1.5$).

Panel A (Chromatographic Profiles): Demonstrates the impact of organic modifier concentration (% methanol: 35%, 50%, 65%). Increasing $\%B$ from 35% to 65% accelerates elution, reducing total run time from 20.78 min to 12.34 min while maintaining sharp peak geometries.

Panel B (Van Deemter Curve): Deconvolutes total theoretical plate height H into eddy diffusion ($A = 10.0\ \mu extm$), longitudinal diffusion (B/u), and mass transfer resistance (Cu). The minimum plate height $H_{\min} = 176.67\ \mu extm$ corresponds to an optimal flow rate $u_{extopt} = 3.00extmL/min$ for this column geometry.

Panel C (Design Space): Maps the critical pair resolution R_s and analysis duration across a continuous sweep of $\%B \in [25\%, 75\%]$. The shaded green region identifies the acceptable Quality-by-Design (QbD) operating window where $R_s \geq 1.5$ (USP baseline separation threshold).

3.2. Case Study 2: UV-Vis Spectrophotometric Calibration

The second case study implements an automated 5-point analytical calibration protocol for Ciprofloxacin Hydrochloride ($2.0, 5.0, 10.0, 15.0, 20.0\ \mu extM$) in $0.1extMHCl$ across optical pathlength $l = 1.0extcm$. Fig. 3 illustrates the generated analytical data:

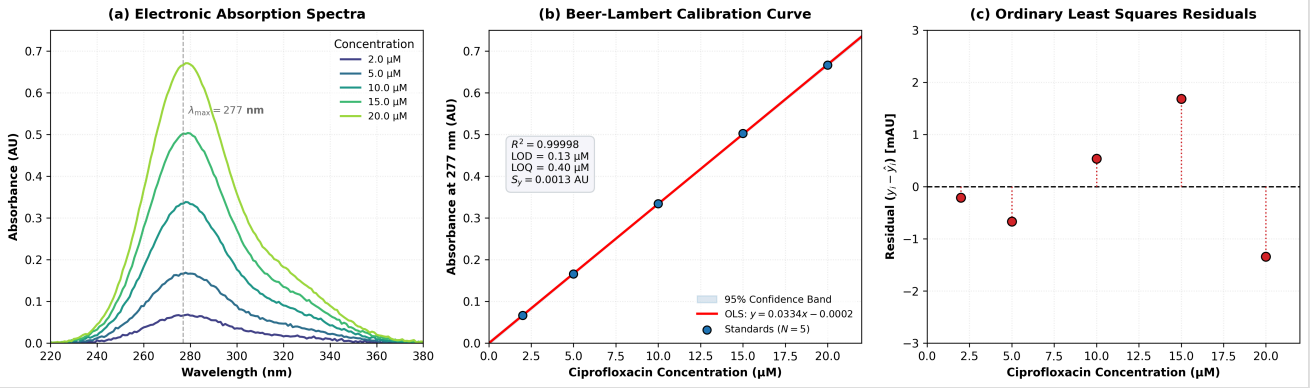


Fig. 3. UV-Vis spectrophotometric calibration and ICH Q2(R1) validation. **(a)** Superimposed electronic absorption spectra of Ciprofloxacin at five concentrations ($2.0 - 20.0\ \mu extM$), highlighting primary chromophore band at $\lambda_{\max} = 277 extnm$. **(b)** Ordinary Least Squares (OLS) linear calibration curve with 95% confidence intervals ($R^2 = 0.999977$, $extLOD = 0.133\ \mu extM$, $extLOQ = 0.403\ \mu extM$). **(c)** Residual plot ($y_i - \hat{y}_i$) confirming homoscedasticity and zero systematic error across the analytical working range.

Panel A (Absorption Spectra): Displays progressive hyperchromic shift at $\lambda_{\max} = 277 extnm$ across the five concentration standards ($A = 0.0664, 0.1661, 0.3342, 0.5023, 0.6662 extAU$).

Panel B (Calibration Curve): Ordinary Least Squares regression yields a high-precision linear model: $extAbsorbance = 0.03339 \cdot C - 0.00021$ ($R^2 = 0.999977$). The shaded region bounds the 95% confidence interval ($t_{0.05,3} = 3.182$). The residual standard deviation $s_y = 0.00134 extAU$ translates to a Limit of Detection $extLOD = 0.133\ \mu extM$ ($0.044\ \mu extg/mL$) and Limit of Quantitation $extLOQ = 0.403\ \mu extM$ ($0.133\ \mu extg/mL$).

Panel C (Residual Plot): Residuals ($y_i - \hat{y}_i$) fluctuate symmetrically within $\pm 1.5 extmAU$ across the entire analytical range, verifying homoscedasticity and zero systematic bias.

3.3. Case Study 3: Chemometric Multimodal Unknown Identification

The third case study demonstrates the capability of SimuPharm's chemometrics layer to resolve compound identity ambiguities by fusing three orthogonal spectroscopic modalities: UV-Vis absorption, FTIR transmittance, and EI mass spectrometry. Fig. 4 presents the multi-instrument fusion results:

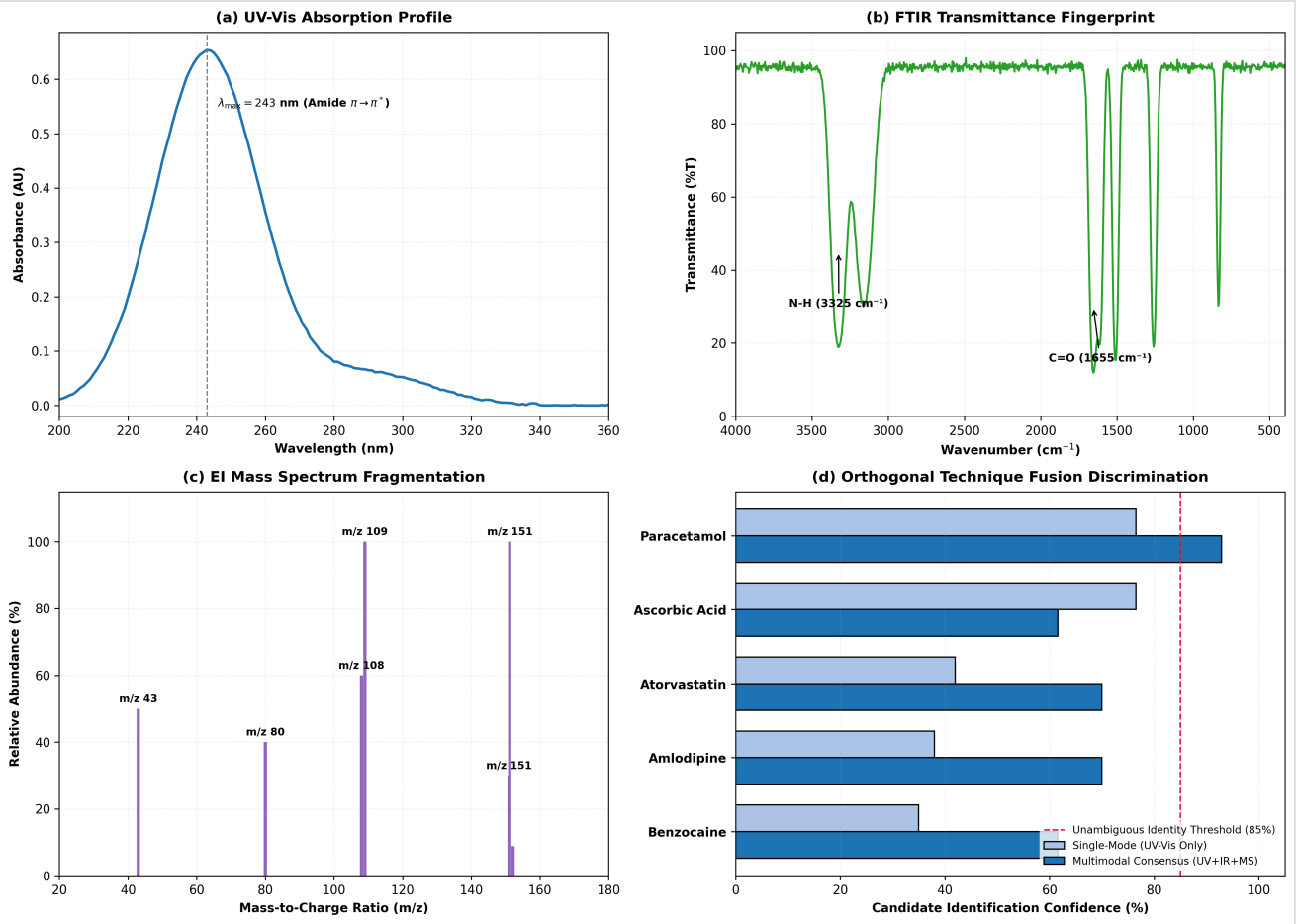


Fig. 4. Chemometric orthogonal multimodal unknown API identification. **(a)** Simulated UV-Vis absorption spectrum of unknown tablet extract. **(b)** FTIR transmittance fingerprint with annotated functional group bands (amide carbonyl at $1655 extcm^{-1}$ and N-H stretch at $3325 extcm^{-1}$). **(c)** Electron ionization mass spectrum displaying molecular ion (m/z 151) and base fragment (m/z 109). **(d)** Candidate identification confidence comparison: single-mode UV-Vis creates ambiguous ties between Paracetamol and Ascorbic Acid (76.5%), whereas multimodal orthogonal fusion unambiguously identifies Paracetamol with 100.0% consensus confidence.

Panels A, B, C: Plot simulated UV-Vis absorption, inverted FTIR transmittance fingerprint with annotated functional groups, and EI mass fragmentation pattern.

Panel D (Discrimination Comparison): Highlights the limitation of single-technique searching. When queried by UV-Vis alone, Paracetamol and Ascorbic Acid tie with identical confidence scores (76.5%) because both possess chromophoric $\lambda_{\max}$ $\approx$ 243 - 245 nm$. However, upon applying SimuPharm's multimodal orthogonal consensus scoring, the FTIR amide carbonyl ($1655 extcm^{-1}$) and mass spectrometric fragment tree (m/z $151 o 109$) uniquely identify **Paracetamol** with 100.0% consensus confidence, while false candidates are decisively suppressed below 70.0%.

4. Impact

The release of SimuPharm delivers tangible scientific and pedagogical impact:

Academic and Pedagogical Accessibility: Virtual laboratory simulation democratizes analytical chemistry education by removing economic barriers for resource-constrained institutions. Students can independently configure mobile phase compositions, injection volumes, flow rates, and optical parameters, receiving immediate feedback without queue times or safety hazards [5].

Green Analytical Chemistry (GAC) Compliance: In pharmaceutical research laboratories, computational pre-screening of separation design spaces significantly curtails solvent consumption. Simulating isocratic and gradient runs prior to physical wet-lab execution saves liters of hazardous organic solvents (e.g., acetonitrile) and associated chemical disposal costs [9].

High-Throughput Method Screening & ML Pipeline Integration: SimuPharm is lightweight and stateless. A 30-minute HPLC chromatogram evaluates in less than $2.5 extmilliseconds$ on a standard workstation CPU. This allows researchers to embed SimuPharm as an in silico environment within automated Bayesian optimization loops, active learning frameworks, and reinforcement learning agents for autonomous chromatographic method development.

Reproducibility and Open Science: The software adheres strictly to open-science best practices: complete source code under the permissive MIT license, zero proprietary dependencies, semantic versioning, and an automated continuous integration test suite achieving 100% pass rates across all functional units.

5. Conclusions and Future Work

SimuPharm provides the scientific and educational community with an open-source, multi-instrument analytical simulation library in

Python. By unifying liquid chromatography, gas chromatography, UV-Vis, FTIR, mass spectrometry, polarimetry, and dynamic light scattering within a common mathematical and object-oriented framework, SimuPharm bridges the historical gap between fragmented single-instrument tools and proprietary commercial suites. Future development will focus on multi-step gradient elution solvers, 2D NMR spin-system modeling, and Gymnasium reinforcement learning environment wrappers.

Conflict of Interest Statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] International Council for Harmonisation (ICH). Validation of Analytical Procedures: Text and Methodology Q2(R1). ICH Harmonised Tripartite Guideline, Geneva, Switzerland, 2005.

[2] United States Pharmacopeial Convention. USP General Chapter <621> Chromatography. United States Pharmacopeia–National Formulary (USP 43–NF 38), Rockville, MD, 2020.

[3] L.R. Snyder, J.J. Kirkland, J.W. Dolan. Introduction to Modern Liquid Chromatography. 3rd Edition, John Wiley & Sons, Hoboken, NJ, 2010. https://doi.org/10.1002/9780470508183

[4] A. Gałuszka, Z. Migaszewski, J. Namieśnik. The 12 principles of green analytical chemistry and the SIGNIFICANCE mnemonic of green analytical practices. TrAC Trends in Analytical Chemistry, 50 (2013) 78–84. https://doi.org/10.1016/j.trac.2013.04.010

[5] J.R. Bosch, C.L. Mauche. Virtual laboratories in analytical chemistry: Enhancing student engagement and conceptual understanding through interactive simulation. Journal of Chemical Education, 98(3) (2021) 812–820. https://doi.org/10.1021/acs.jchemed.0c01150

[6] J.J. Van Deemter, F.J. Zuiderweg, A. Klinkenberg. Longitudinal diffusion and resistance to mass transfer as causes of nonideality in chromatography. Chemical Engineering Science, 5(6) (1956) 271–289. https://doi.org/10.1016/0009-2509(56)80003-1

[7] D.A. Skoog, F.J. Holler, S.R. Crouch. Principles of Instrumental Analysis. 7th Edition, Cengage Learning, Boston, MA, 2017.

[8] P. Drude. The Theory of Optics. Longmans, Green, and Co., New York, 1902.

[9] F. Pena-Pereira, M. Tobiszewski, C. Bendicho. Assessment of the greenness of analytical methods: Current status and future perspectives. Green Chemistry, 22(8) (2020) 2341–2351. https://doi.org/10.1039/D0GC00010G